

Motion-Consistent Memory for Foundation Model-Based Tumor Tracking in MR-guided radiotherapy

Ornela Megne¹, Marcel Nachbar², Aya Ghouli¹, Cihan Gani², Daniela Thorwarth², and Thomas Kuestner¹

¹Medical Image and Data Analysis (MIDAS.lab), Department of Diagnostic and Interventional Radiology, University Hospital of Tuebingen, Germany

²Department of Radiation Oncology, University Hospital of Tübingen, Germany

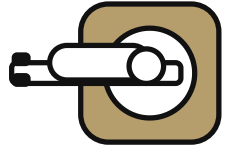
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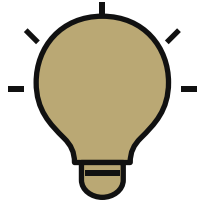
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Motivation



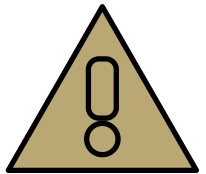
MR-guided radiotherapy relies on real-time tumor tracking

Tracking output influence beam delivery



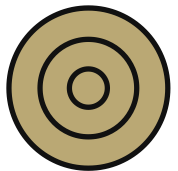
Why it matters

- Tracking failure: geometric miss of the target
- Prolonged failures compromises treatment safety



Challenges

- Respiratory motion
- Low-soft tissue contrast
- Imaging artifacts

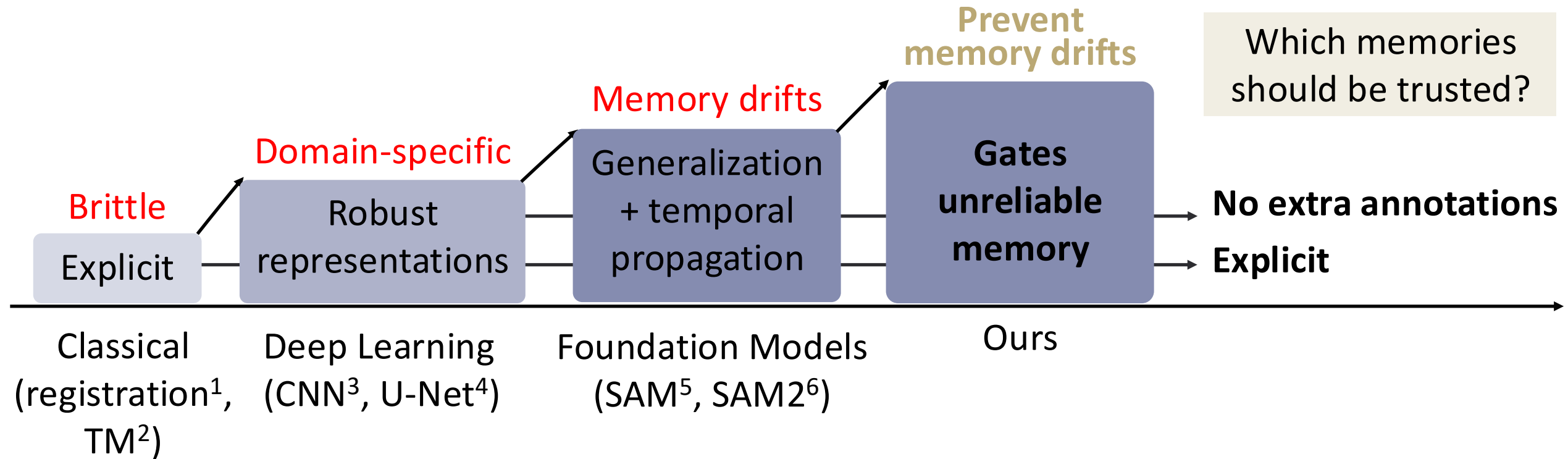


Goal

Robust, real-time tumor tracking that holds up through motion

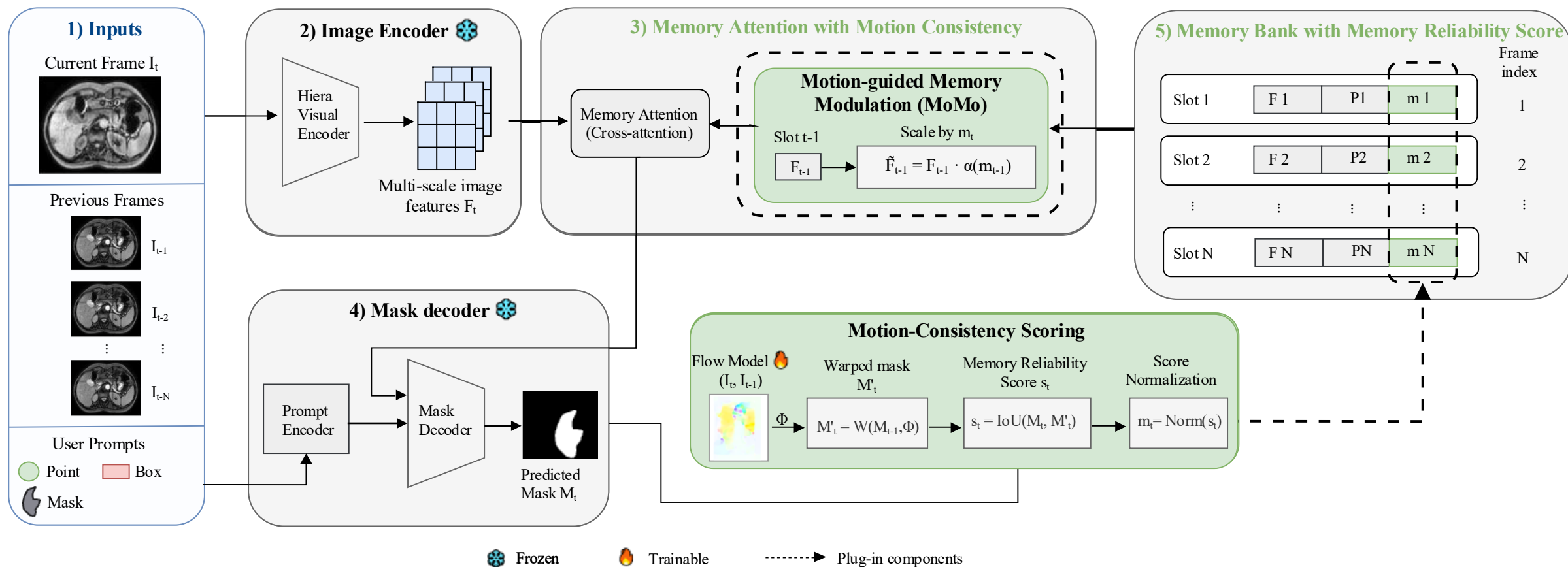


Background

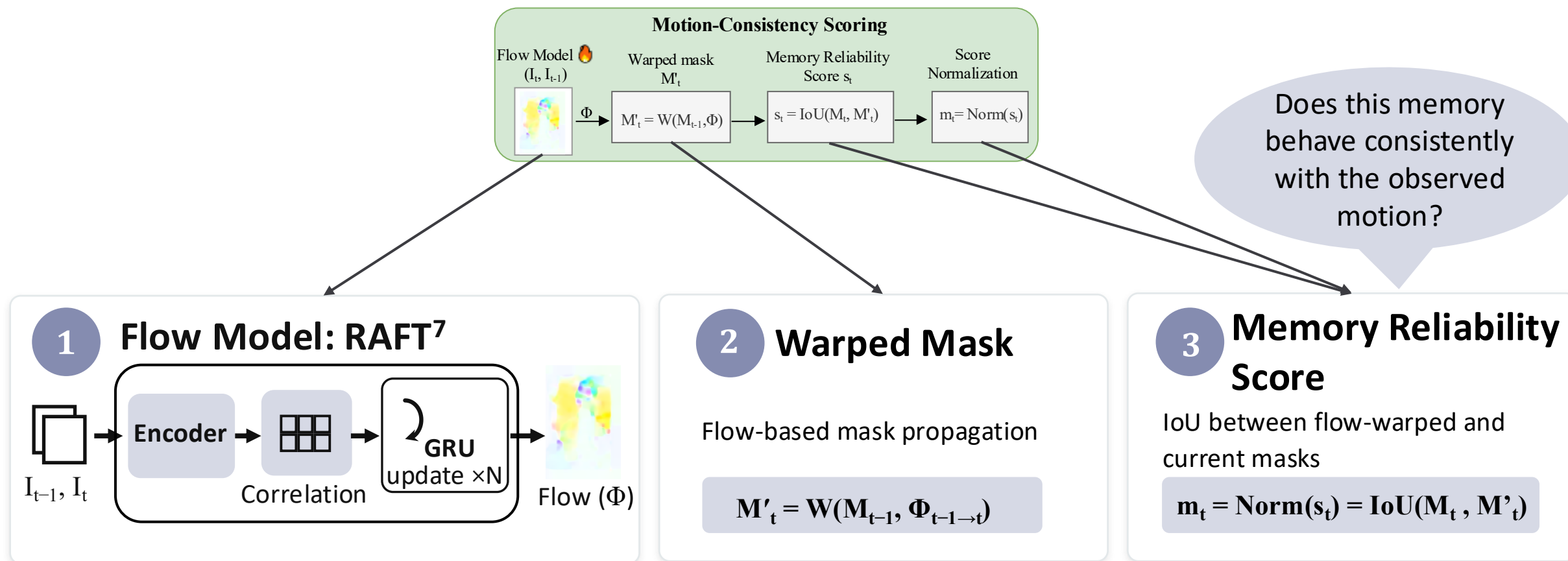


Hypothesis: Motion guidance can provide a complementary signal to identify and suppress unreliable memories.

Method

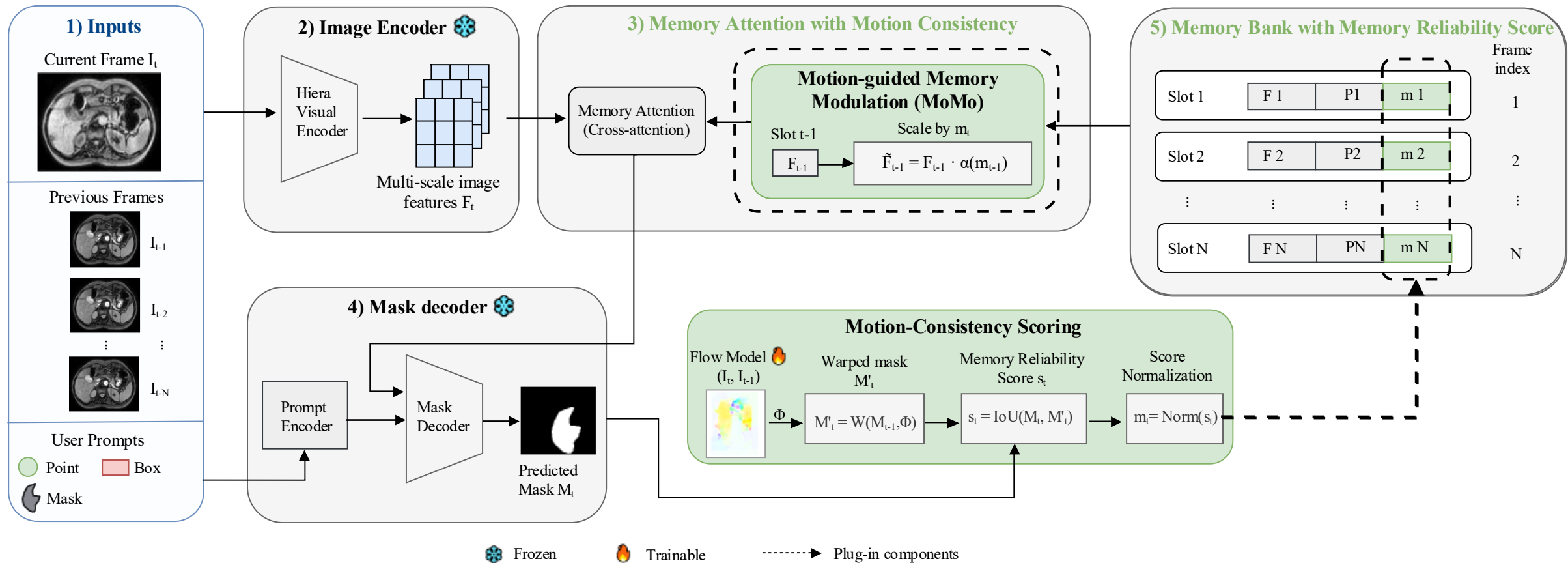


Vanilla SAM2 + MoMo

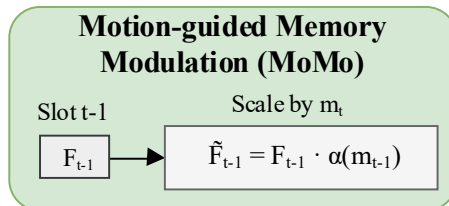


RAFT: Recurrent All-Pairs Field Transforms

Method



Vanilla SAM2 + MoMo



4 MoMo

Gate stored features

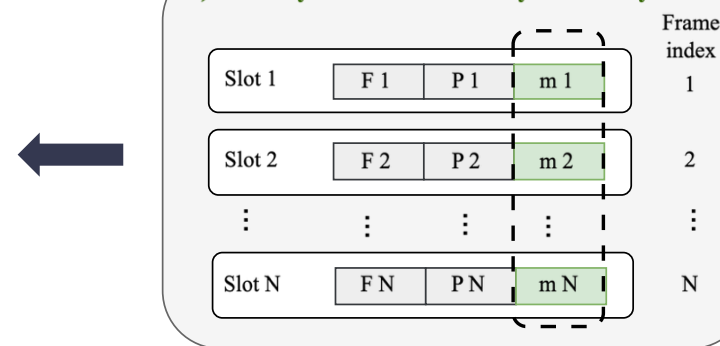
$$\tilde{F}_{t-1} = F_{t-1} \cdot \alpha(m_{t-1})$$



Preserve reliable memories and attenuate unreliable ones

$$\alpha(m_{t-1}) = \begin{cases} 1 & \text{if } m_{t-1} > \tau_1 \\ m_{t-1} & \text{if } m_{t-1} \leq \tau_1 \end{cases}$$

5) Memory Bank with Memory Reliability Score



Experimental Setup

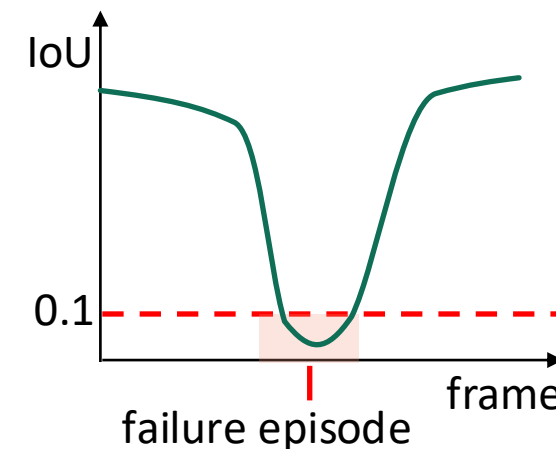


TrackRAD2025 Challenge Dataset⁸: 2D sagittal cine MRI on 0.35T MRIdian and 1.5T Unity MR-Linac with 525 training cases and 30 final test cases

Methods	Parameters
Flow model: Finetuned RAFT	Losses: NMI ¹⁰ +Photometric ¹¹ +smoothness ¹²
Baselines: Frozen SAM2 ¹ and MedSAM2 ⁹	Bounding box + click
Framework: MoMo	Reliability threshold $\in [0.2, 0.3]$

Evaluation metrics

- Segmentation quality: Dice, HD95, J score, F score, J&F, IoU
- Detection rate at IoU thresholds of 0.1 (Det@0.1)
- Temporal stability (Temp-IoU)
- Failure episode (missed frames, miss rate, mean duration, max failure duration in frames)



Results



Method	Dice (%) ↑	HD95 (pixels) ↓	J&F (%) ↑	Det@0.1 (%) ↑	Temp- IoU↑	Time (ms) ↓
SAM2-S	87.90	4.80	85.04	96.0	81.24	169.5
SAM2-MoMo	89.21	4.46	86.06	97.3	80.70	191.4
MedSAM2-S	86.80	4.82	80.54	89.9	82.36	45.1
MedSAM2-MoMo	86.55	5.09	83.84	93.1	82.83	65.1

+1.54%

Dice vs. SAM2-S

+3.94%

J&F vs. MedSAM2-S

< 200 ms

per frame, all variants

➤ *MoMo improves SAM2 with Dice, HD95, J&F and shows a small trade-off for MedSAM2*

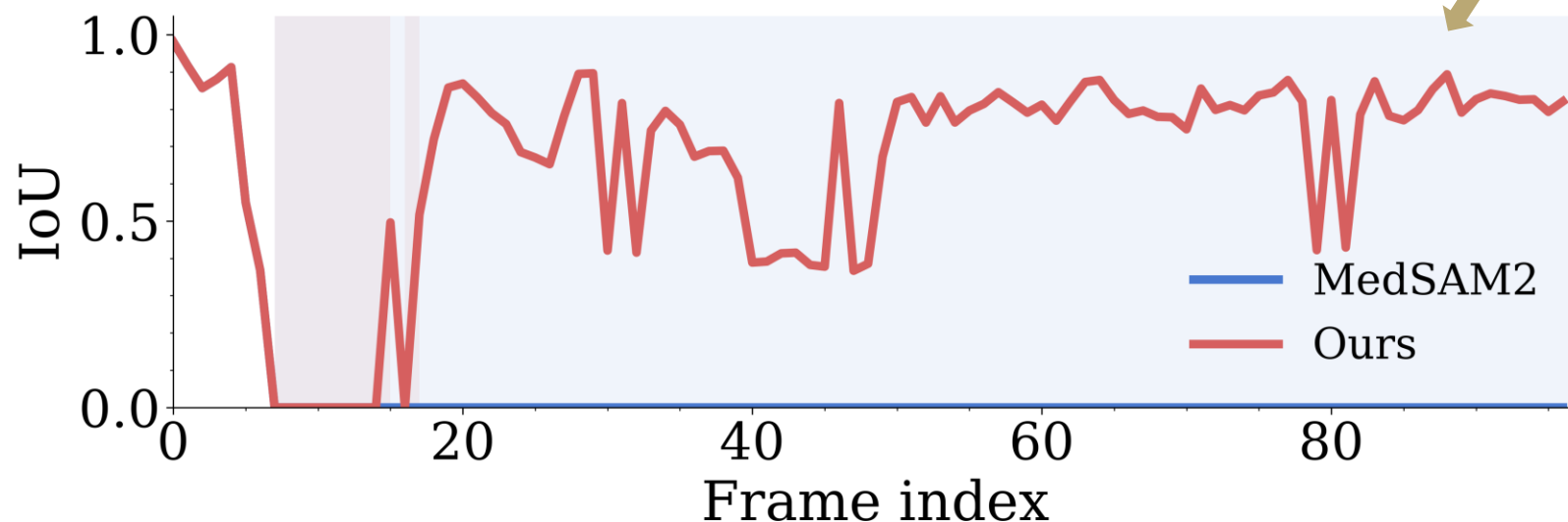
Results



Method	Missed frames↓	Miss rate↓	Mean dur [frames] ↓	Max dur [frames] ↓
SAM2-S	101	4.5%	14.4	91
SAM2-MoMo	75	3.3%	5.0	15
MedSAM2-S	138	6.1%	10.6	90
MedSAM2-MoMo	29	1.3%	2.6	8

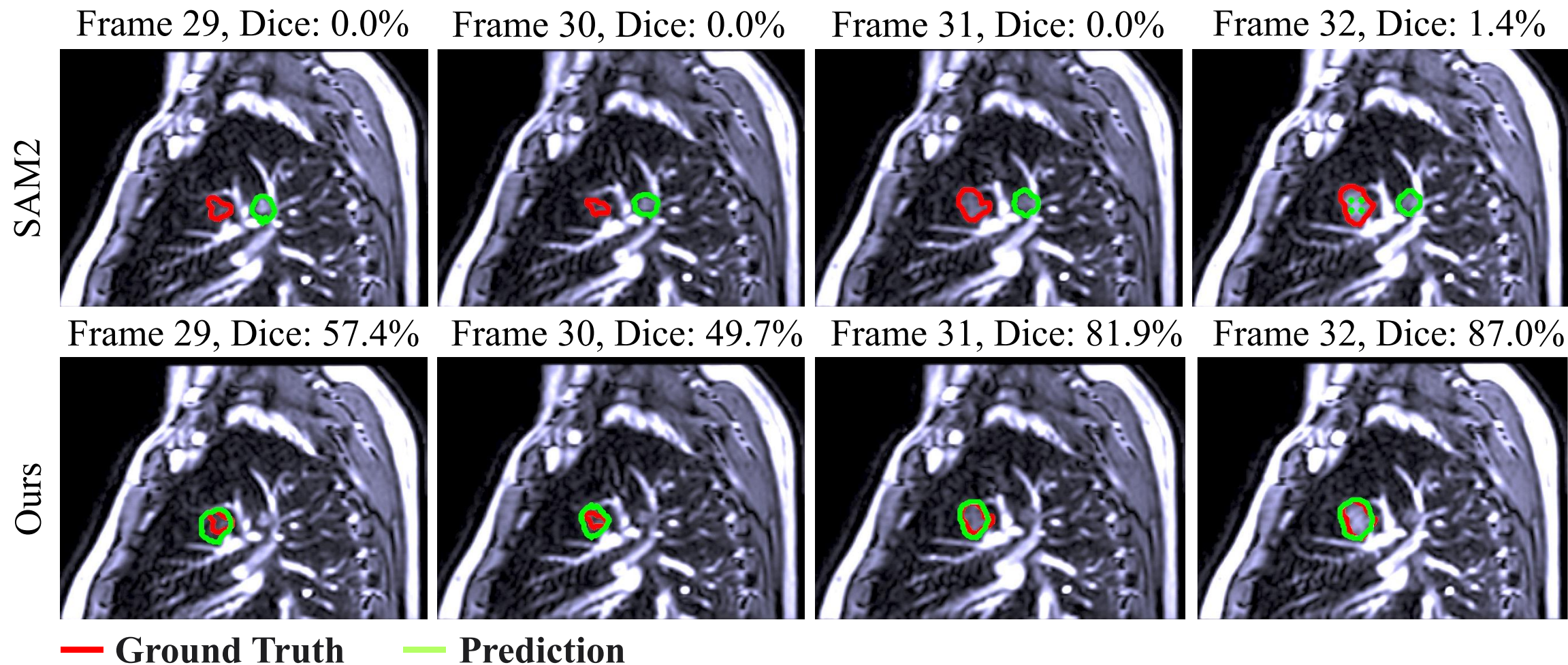
**+109 recovered frames,
miss rate 6.1% → 1.3%
vs. MedSAM2-S**

**8 frames (~2s at 4 Hz) of
max failure duration
vs. MedSAM2-S**



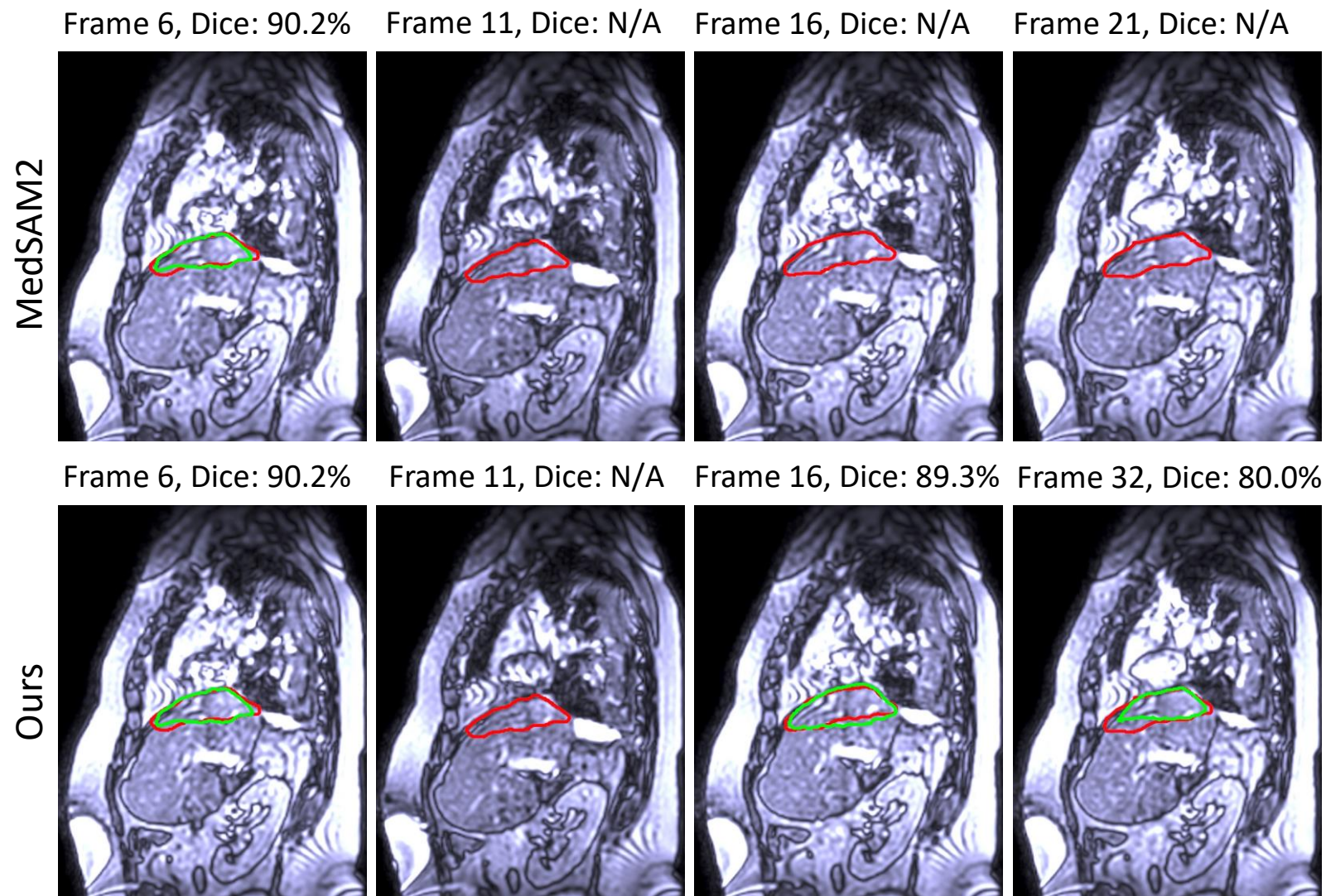
➤ *MoMo reduces sustained tracking failures*

Results



➤ *MoMo gates unreliable memory and recovers the correct target*

Results



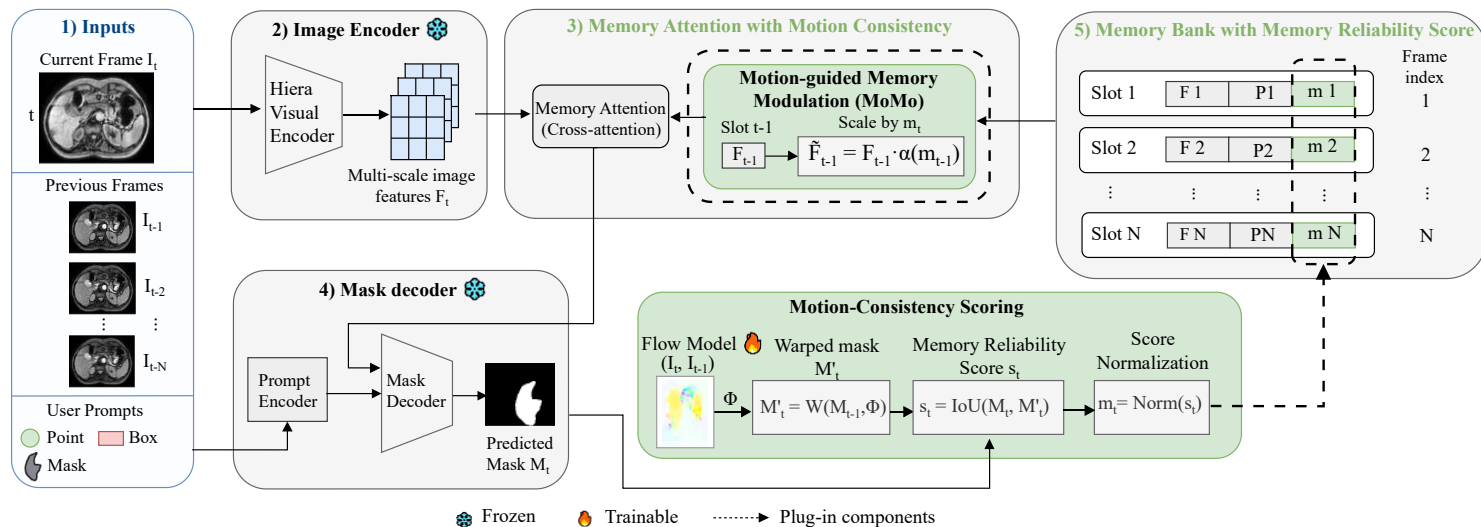
Conclusion



MoMo: Motion-guided memory Modulation

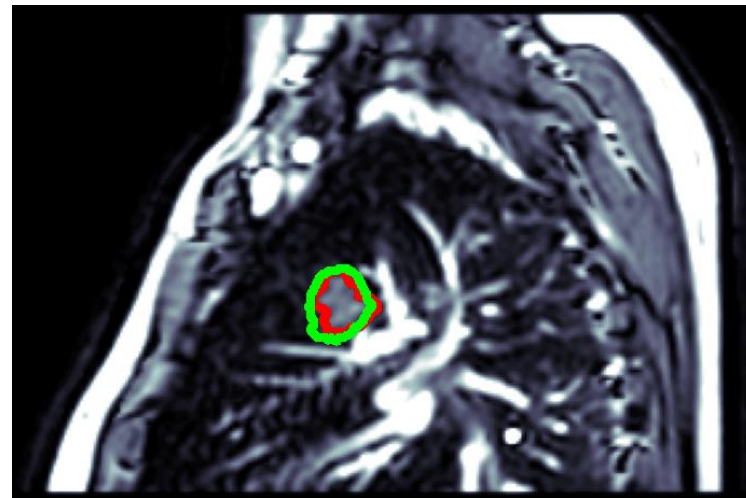
- Lightweight & plug-and-play mechanism
- Gates unreliable memory on retrieval
- Reduces prolonged tracking failures
- Stays below 200ms/frame

Future works: domain generalization, larger prospective validation



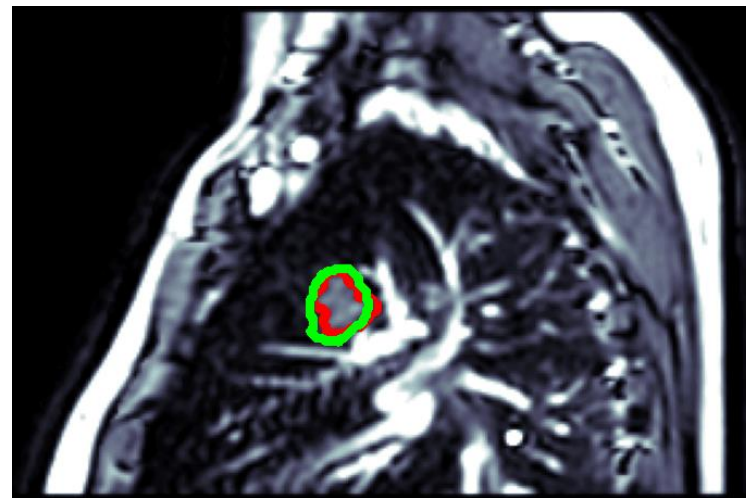
Frame 20, Dice: 0.801

SAM2



Frame 20, Dice: 0.797

SAM2-MoMo



Thank you very much for your attention!

Acknowledgements



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